

TEAM ASTROGG

Amiya Upendra Natarajan - 98848 33375

Ananya Shriram - 95996 64788

Vedh Sontha - 99029 05189

*Messier 45, the Pleiades
Taurus*

PROBLEM STATEMENT 1

AUTONOMOUS EXOPLANET DETECTION FROM LIGHT CURVES

BUILD A PIPELINE THAT TAKES RAW KEPLER/TESS LIGHT CURVES, CLEANS OUT NOISE AND STELLAR VARIABILITY, AND FINDS REPEATING DIPS THAT POINT TO A PLANET CROSSING ITS STAR. A TRAINED MODEL CAN THEN DECIDE WHICH OF THOSE SIGNALS ARE REAL PLANETS, AND A DASHBOARD DISPLAYS THE CANDIDATES WITH CONFIDENCE SCORES, INCLUDING STARS THAT HOST MORE THAN ONE PLANET.

MOTIVATION AND RELEVANCE TO ASTRONOMY

Raw Kepler / TESS light curves are noisy due to

stellar variability, instrument drift, and cosmic-ray hits

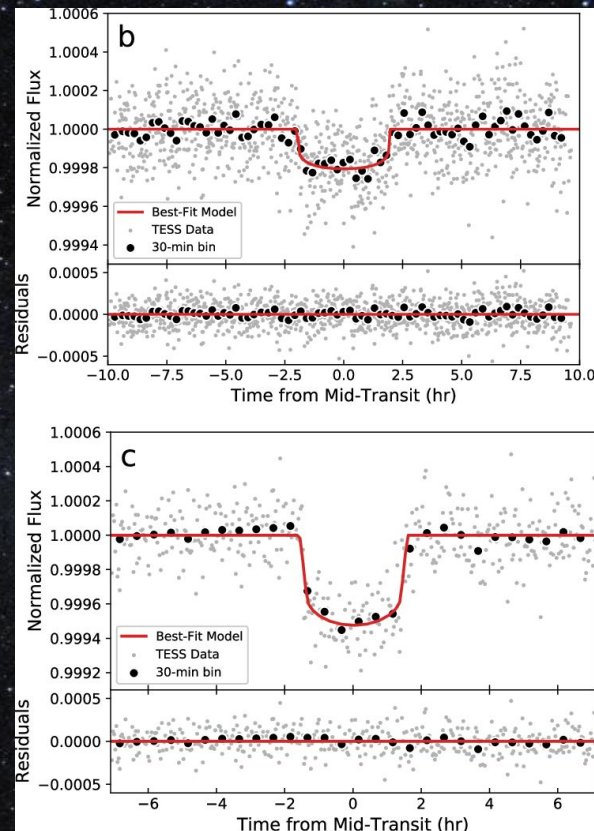
for an Earth-sized planet around a Sun-like star, it's close to 0.01%; practically a needle in a haystack.

raw data-generation rate has outpaced the ability of astronomers to manually review candidates

these transit surveys would be used for identifying Earth-sized planets in the habitable zone of their stars

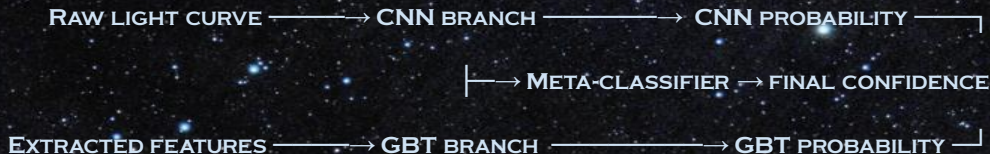
The goal is to build a pipeline that autonomously cleans each curve, finds repeating transit-like dips, and uses a trained model to decide which are real planets, and flagging stars that host more than one.

A star can dim by less than 0.01% during transit



Kane, Stephen R., et al. "Transits of known planets orbiting a naked-eye star." *The Astronomical Journal* 160.3 (2020): 129.

PROPOSED STACK



1.

CURATED DATA

Kepler / TESS light curves pulled via *lightcurve* (wraps MAST archive access) and NASA's PDCSAP flux pipeline for preprocessing.

NASA Exoplanet Archive – Cumulative KOI (Kepler Object of Interest) Table (Contains labeled dispositions: CONFIRMED, CANDIDATE, FALSE POSITIVE)

2.

DETREND & CLEAN

Remove stellar variability and instrumental trends while preserving short-duration transit dips – sigma-clip outliers first, then detrend with wotan's biweight filter, and normalize flux.

3.

TRANSIT SEARCH

Box Least Squares finds the strongest periodic dip; mask it and re-search the residual 2-3 times to recover additional planets.

4.

ENSEMBLE CLASSIFICATION

1D CNN learns transit shape directly from the phase-folded light curve, gradient-boosted tree model (XGBoost) on hand-engineered physical features, and a lightweight meta-classifier that blends both models' outputs into a final confidence score

5.

DASHBOARD

A Streamlit app displays candidates, phase-folded plots, confidence scores, and multi-planet flags live.

TECHNOLOGY STACK

DATA & ASTRONOMY LIBRARIES

MAST Archive
NASA Exoplanet Archive
*lightkurve, wotan,
astropy*

MACHINE LEARNING

TensorFlow / Keras (1D
CNN)
xgboost, scikit-learn
numpy, pandas

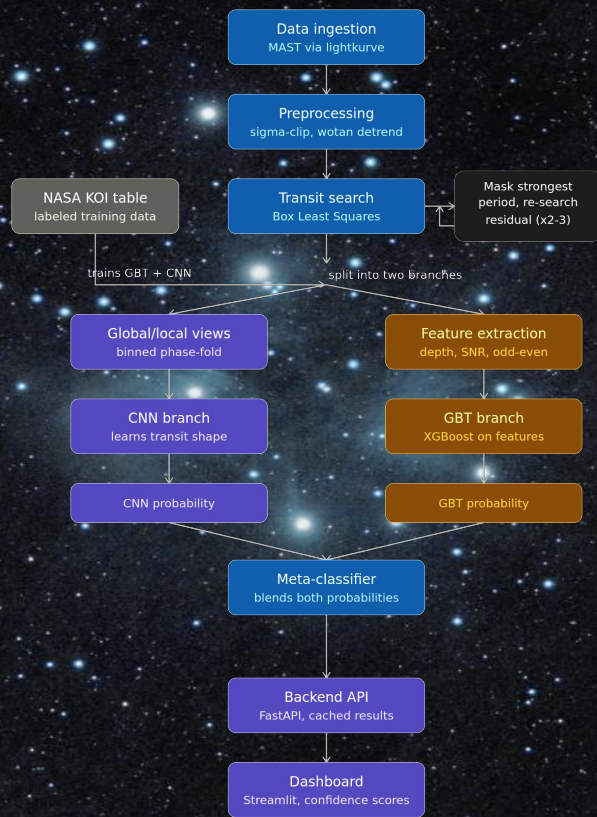
FRONTEND

Streamlit
Plotly

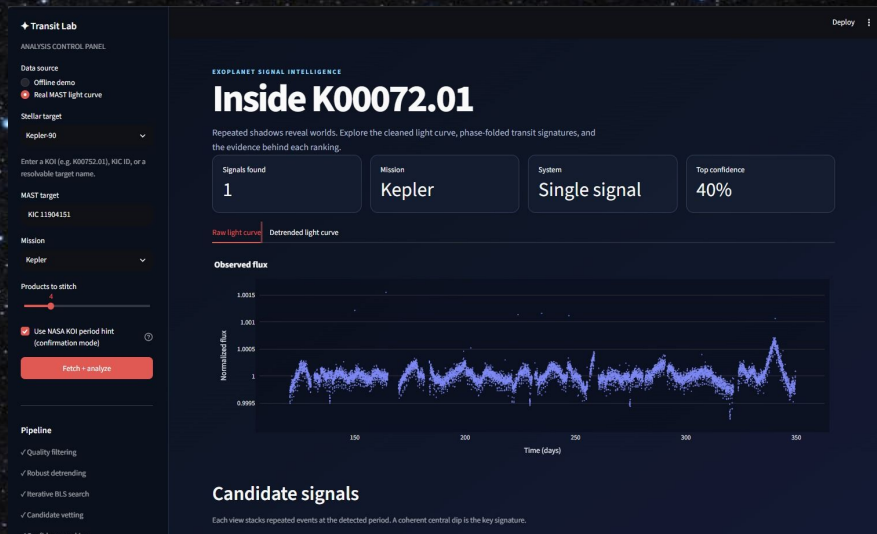
BACKEND

Python
FastAPI

ARCHITECTURE FLOW DIAGRAM



Real MAST Prediction Result (ONLY xgboost)



But the classifier assigned a 40% planet-likeness score below the 43.1% threshold, so it was classified as likely a false positive.

We analyzed four real Kepler light-curve products from MAST using *lightcurve*. The detected 0.837-day signal had 164 ppm depth, SNR 64.5, and 253 observed transits.



FUTURE SCOPE AND REAL WORLD APPLICATIONS

Real-Time TESS Alerts

Extend the pipeline to ingest new TESS sectors as they're released and flag candidates automatically.

Citizen Science Integration

Feed high-confidence candidates into platforms like Planet Hunters for community-driven vetting.

Next-Gen Missions

Adapt the detection and classification stages for PLATO and other upcoming survey missions.

Follow-Up Prioritization

Rank candidates by confidence to guide limited ground-based telescope time toward the best targets.

Deeper Characterization

Extend the classifier to estimate planet radius and orbital period, not just real-vs-false-positive.

Transfer to Other Surveys

Retrain the same pipeline architecture on other time-series survey data beyond exoplanet transits.